

High Catalytic Activity in Upgrading of Ethanol to Aromatic Alcohols over Zinc Hydroxyapatite

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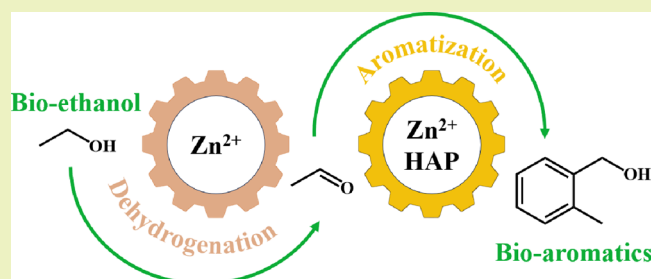
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ABSTRACT: The direct conversion of ethanol to aromatic alcohols has attracted much attention as increasingly available biomass ethanol fermented from renewable sources. Herein, we present a highly efficient zinc hydroxyapatite for direct synthesis of aromatic alcohols from ethanol with a specific rate of 1.95×10^{-7} mol_{aromatics} g_{catal}⁻¹ s⁻¹, which is up to now the best in the literature. The γ -C of 2-butenal after deprotonation undergoes the nucleophilic attack to the carbonyl-C of another 2-butenal to form 2,4,6-octatrienal, which is subsequently cyclized to give ortho-aromatics, as the main pathway for the aromatization. Dynamic and theoretical studies reveal that the Zn site plays a role in hydrogen transfer for dehydrogenation and hydrogenation. The substitution of the Zn atom for the Ca atom was identified at the strong basic [Ca–O–P] site, and the Zn site lowers the energy barrier to ethanol dehydrogenation. The rate-determining step of this reaction is the coupling of two acetaldehyde molecules to form 2-butenal. This work provides a strategy to catalyze the efficient conversion of ethanol to high value-added aromatic alcohols and contributes to the development of protocols for sustainable synthesis of aromatic alcohols from biomass through heterogeneous catalysts.

KEYWORDS: ethanol upgrading, aromatic alcohol, hydroxyapatite, zinc, DFT



1. INTRODUCTION

Biomass-derived feedstocks offer promising avenues for sustainable production of green fuels and chemicals,¹ with ethanol being a prominent candidate due to its abundant availability.^{2,3} Ethanol can serve as a valuable platform molecule for the synthesis of high value-added chemicals,^{4–6} including methylbenzyl alcohol (MB), which finds applications as an important organic chemical intermediate in pharmaceuticals and agrochemicals.^{7,8} Traditionally, methylbenzyl alcohols are mainly produced by the oxidation of petroleum-derived xylene in a pressurized dioxygen atmosphere.⁹ Under such harsh oxidative conditions, target products can be easily oxidized further, suffering from low production efficiency and environmental concerns for product separation.¹⁰ Therefore, exploring nonpetroleum routes using sustainable feedstocks is essential to develop environmentally friendly and cost-effective methodologies.

Since the upgrading of ethanol envelops complex reaction channels competitive against each other, it is challenging to use ethanol as a feedstock for the direct production of methylbenzyl alcohol. Thus, only a few studies considered the conversion from ethanol or acetaldehyde to methylbenzyl alcohol. Resasco et al. and Flaherty et al. reported the synthesis of methyl benzaldehyde from acetaldehyde on magnesium oxide and hydroxyapatite, respectively, with a yield of less than 13% for the target aromatic product,^{11,12} which was attributed

to the nonselective intermolecular hydrogen transfer between the enal intermediate and ethanol that resulted in aliphatic alcohols as the main product. Lusardi et al. reported the upgrading of acetaldehyde to tolualdehydes catalyzed by Al-doped MgO (MgAl₂O₄) with a selectivity of less than 10% for para- and ortho-aromatic aldehydes and argued that the promotion of para-aromatic aldehydes compared to ortho-aromatic aldehydes was the consequence of redistributed atomic charge of 2-butenal intermediate on the defective composite MgAl₂O₄.¹³

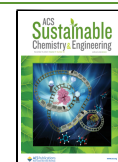
Hydroxyapatite (HAP) has been reported to show activity to catalyze an ethanol coupling reaction with high selectivity for butanol, which is attributed to the presence of adjacent Lewis acid–base pairs.^{14,15} Although HAP has been often used for upgrading of ethanol through C–C bond coupling, it is inert for dehydrogenation, which is however necessary to produce unsaturated products, e.g., aromatics.¹² There are studies showing that upon the replacement of the Ca²⁺ ions on the surface of hydroxyapatite by other metal cations, such as

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cobalt,¹⁶ nickel,¹⁷ and zinc,¹⁸ HAP can display reactivity to promote a hydrogen transfer reaction. Recently, we reported the use of bifunctional cobalt hydroxyapatite catalysts for the direct conversion of ethanol to methylbenzyl alcohol, and benzyl alcohol can also be obtained via methanol–ethanol cofeeding.^{19,20} This catalyst shows high selectivity for aromatic products but with an ethanol conversion of about 30% under optimal conditions and is proposed to promote the hydrogen transfer at the transition metal dopant sites and the C–C bond coupling on the Ca^{2+} sites of HAP. To achieve high conversion of ethanol, in this work, we prepared a Zn-doped HAP catalyst to enhance the ability of the catalyst to dehydrogenate the substrate, and the catalytic test indicated that the designed Zn-doped catalyst exhibited significantly improved catalytic activity and promoted the conversion of renewable feedstock ethanol to aromatic products through a dehydrocyclization mechanism. This work represents a step toward sustainable and environmentally friendly processes for the production of aromatic chemicals, and it demonstrates the potential of using Zn-doped hydroxyapatite as a catalyst for efficient conversion of ethanol to methylbenzyl alcohol. The findings contribute to the advancement of green chemistry practices, paving the way for the utilization of biomass-derived feedstocks in the synthesis of high value-added chemicals.

2. EXPERIMENTAL SECTION

2.1. Catalyst Synthesis. Hydroxyapatite (HAP) obtained from Acros Organics with a Ca/P ratio of 1.67 was used as the precursor for the synthesis of zinc-loaded hydroxyapatite (Zn-HAP) through an impregnation method. In this method, HAP was impregnated with $\text{Zn}(\text{NO}_3)_2$ (Sinopharm) and deionized water at room temperature for 0.5 h. Subsequently, the impregnated samples were dried in air at 50 °C for 12 h and then calcined in air at 400 °C for 2 h at a rate of 5 °C min^{-1} to obtain the as-prepared Zn-HAP catalyst.

2.2. Characterization. Powder X-ray diffraction (XRD) was conducted on a Panalytical X'pert Pro Super X-ray diffractometer using Cu K α radiation (40 kV, 40 mA, $\lambda = 0.15406$ nm) over an angular range of 5–80° (2 theta). Inductively coupled plasma optical emission spectrometry (ICP-OES) was used to characterize the zinc content by an Optima2000DV. Scanning transmission electron microscopy (STEM) images were obtained using a Hitachi HF5000 instrument at 200 kV. The strength and number of acid and base sites on the surfaces were studied by temperature-programmed desorption with mass spectrometry (TPD-MS) using a Micromeritics AutoChem II 2920 instrument. Typically, 100 mg of sample was loaded into a U-shaped quartz tube and treated in a He flow at 400 °C for 2 h to clean the surface. When the sample temperature dropped to 100 °C, the surface was sequentially swept by a He flow for 0.5 h, a CO_2 or NH_3 flow for 0.5 h, and then a He flow for 1 h to remove any weakly adsorbed CO_2 . Finally, the sample was heated from 50 to 400 °C in a He flow with a rate of 10 °C min^{-1} .

2.3. Catalytic Activity Measurements. The catalytic performance was tested under atmospheric pressure in a quartz-tube fixed bed microreactor with an inner diameter of 8 mm. The catalyst was pretreated in the N_2 flow at 400 °C for 2 h before the test, and the same bed height was maintained. During the test, the flow rate of the carrier gas nitrogen is 30 $\text{mL} \cdot \text{min}^{-1}$ to ensure that the catalytic reaction is in the kinetic zone. Ethanol, dried prior to use by mixing with calcium oxide to remove water, was fed into the reactor through a bubbler with a N_2 carrier. The gas product was analyzed using a gas chromatograph (GC) with a flame ionization detector (FID) and a thermal conductivity detector (TCD). The Zn-HAP catalyst exhibited good stability in a 40 h test, showing an average carbon deposition rate of 0.0035 mg/min. As the feed rate of ethanol is 3.7 mg/min, the deposited carbon only accounts for 0.1%, indicating that the carbon balance is close to 100%.

The ethanol conversion and product selectivity were calculated on the basis of moles of carbon in the products, respectively:

Ethanol conversion:

$$\text{con. (\%)} = \left(1 - \frac{N_{\text{out,ethanol}} A_{\text{out,ethanol}} f_{\text{out,ethanol}}}{N_{\text{out,ethanol}} A_{\text{out,ethanol}} f_{\text{out,ethanol}} + \sum_{i \geq 2} N_i A_i f_i} \right) \times 100$$

Product selectivity:

$$\text{Sel. (\%)} = \frac{N_i A_i f_i}{\sum_{i \geq 2} N_i A_i f_i} \times 100$$

where N_i , A_i , and f_i represent the carbon number, the FID peak area, and the response factor of each product, respectively.

Turnover frequency (TOF):

$$\text{TOF} = \frac{F_{\text{Ethanol}} \text{Con.}}{M_{\text{Zn}} D_{\text{Zn}}} \text{ (s}^{-1}\text{)}$$

where F_{Ethanol} , M_{Zn} , and D_{Zn} represent the flow rate of ethanol, the molar mass of zinc, and the dispersion of zinc, respectively. The dispersion of zinc is estimated by the XRD patterns.

2.4. Computational Method. All calculations were performed by the density functional theory (DFT) method implemented in the Vienna Ab initio Simulation Package (VASP 5.4.4).^{21,22} The Perdew–Burke–Ernzerhof (PBE) method was adopted for the exchange correlation energy.^{23,24} Hexagonal hydroxyapatite ($(\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$, space group $P6_3/m$) was used to prepare a HAP $2 \times 2 \times 1$ supercell model with its (001) facet exposed to a 15 Å vacuum space.²⁵ The zinc-doped model was prepared by substituting a Zn atom at the Ca_I site (Zn_I -HAP) or Ca_II site (Zn_II -HAP) of HAP (Ca_II represents the site bound with a hydroxyl group and Ca_I represents the site without a hydroxyl group, see in Section 3.3). Single gamma-point grid sampling was used for Brillouin zone integration.²⁶ The cutoff energy of the plane-wave basis set was 450 eV. The bottom half of the models was frozen. The convergence criteria for energy were 1×10^{-5} eV and 0.02 eV $\cdot \text{\AA}^{-1}$ for force. The zero damping DFT-D3 method of Grimme was employed to correct the description of the van der Waals interactions.²⁷ The transition states were obtained by using the DIMER method.²⁸ Frequency calculations were performed to confirm that there was only one imaginary mode for the transition states. The charge distribution was obtained by Bader charge analysis.^{29,30}

The absorption energy (E_{abs}) was determined by

$$E_{\text{abs}} = E(\text{X/surface}) - E(\text{surface}) - E(\text{X})$$

where $E(\text{X/surface})$, $E(\text{surface})$, and $E(\text{X})$ represent the energies of the adsorbed system, the surface, and the isolated adsorbate, respectively.

3. RESULTS AND DISCUSSION

3.1. Catalyst Characterization. The X-ray diffraction of HAP and Zn-HAP was investigated to analyze the effect of Zn doping on their structural properties. Figure 1 shows the diffraction peaks of the (002), (211), and (300) crystal planes on both samples. The angles of these peaks match well with those of the stoichiometric hydroxyapatite used as a reference, identifying that the HAP catalyst presents a hexagonal structure in the $P6_3/m$ space group. The XRD pattern of Zn-HAP retains the features of pure HAP, indicating that the Zn doping did not cause phase transition. No characteristic crystal planes of ZnO , Zn_3P_2 , or $\text{Zn}_3(\text{PO}_4)_2$ were detected at a 1.4 wt % loading of Zn. As the loading of Zn increased, the ZnO peak appeared, and its intensity gradually grew monotonously (Figure S1). This suggests that Zn is highly dispersed at low loadings and anchored at its doping site, which was confirmed

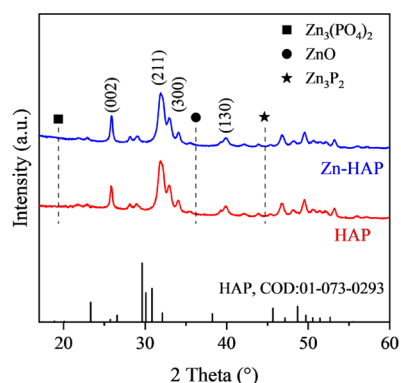


Figure 1. XRD patterns of HAP and Zn-HAP (1.4 wt %).

by the in situ XRD results in the temperature range of 50–600 °C (Figure S2).

The STEM images (Figure 2 and Figure S3) collected from various regions of the catalyst indicated that there were no observable Zn-rich domains, which can be attributed to the high lattice match between Zn and Ca in HAP,^{31,32} as proved by energy-dispersive X-ray spectroscopy (EDS) (Figure 2b). The lack of Zn-containing aggregates on the surface of the Zn-HAP catalyst in the TEM images and the absent peaks of Zn-containing species in the X-ray diffraction pattern indicate the successful preparation of the highly dispersed Zn-HAP catalyst.

The surface basicity of HAP and Zn-HAP, which determines the product distribution,³³ was evaluated by CO₂-TPD experiments. As shown in Figure 3a, three CO₂ desorption peaks at 122, 195, and 253 °C were observed for both catalysts, corresponding to the weak basic sites of the hydroxyl group bound to calcium (marked as Ca–OH), medium basic sites of oxygen bound to two calcium ions (marked as Ca–O–Ca), and strong basic sites of oxygen bound to calcium ion and phosphate (marked as Ca–O–P), respectively.^{15,34} Upon Zn doping, the peak assigned to [Ca–O–P] sites decreased significantly, implicating reduced density of exposed [Ca–O–P] sites on the surface of the Zn-HAP catalyst. In view of the roughly unchanged CO₂ desorption peaks of [Ca–OH] and [Ca–O–Ca] sites, this indicates that the substitution of Zn²⁺ ions for Ca²⁺ ions mainly happened at the [Ca–O–P] sites to generate the [Zn–O–P] sites. No new CO₂ desorption peak was observed in the temperature range of 100–450 °C,

indicating that the generated [Zn–O–P] sites have no or very weak basicity. Furthermore, the acidic sites on the catalyst surface were characterized by NH₃-TPD (Figure 3b). Both show one broad desorption peak in the low temperature range 100–300 °C, indicating that only weak acidic sites were exposed on the catalyst surface. The signal intensity of Zn-HAP is lower than that of HAP, suggesting a smaller number of acidic sites on Zn-HAP.

3.2. Catalytic Performance in Ethanol Upgrading.

3.2.1. Catalytic Performance of Zn-HAP Compared to HAP.

Figure 4 shows the correlation of conversion and product distribution in the temperature range of 250–350 °C. The major difference between the reactions over the HAP and Zn-HAP catalysts was the selectivity of the products with the aliphatic products favored in the former while the aromatic alcohols favored in the latter (Figure 4). On HAP, ethanol mainly underwent coupling reactions to produce straight chain alcohol with over 90% selectivity. At 250 °C, *n*-butanol was the main product. Elevation of temperature resulted in growing hydrocarbon chains, and the major products remained saturated aliphatic alcohols with negligible unsaturated alcohols and aldehydes (Figure 4a).

Compared with HAP, Zn-HAP showed a much higher activity in view of the significantly increased ethanol conversion (29.8% vs 63.1% at 325 °C). In the meantime, the major products were methylbenzyl alcohols, suggesting that the channels to produce aliphatic products were suppressed (Figure 4b) and implicating that the ethanol conversion on HAP and Zn-HAP proceeded along different pathways. Elevation of temperature accelerated the conversion rate significantly but did not change the aromatics distribution, and the selectivity of the ortho- and para-products remained 35 and 2% (±1%). When twice as much ethanol was fed, the selectivity of ortho-aromatics was measured as 32% at 250 °C and 35% at 325 °C, respectively. The marginal difference suggests that the formation of aromatic products reached equilibrium on the Zn-HAP catalyst (Figure S4). At elevated temperatures, the total selectivity of aliphatic alcohols C_{4–12}–OH decreased (20.8%), while that of acetaldehyde increased (23.0% at 325 °C).

We also noted that during the ethanol conversion on Zn-HAP, there was excess amount of acetaldehyde intermediate of up to 23% detected while negligible 2-butenal intermediate,

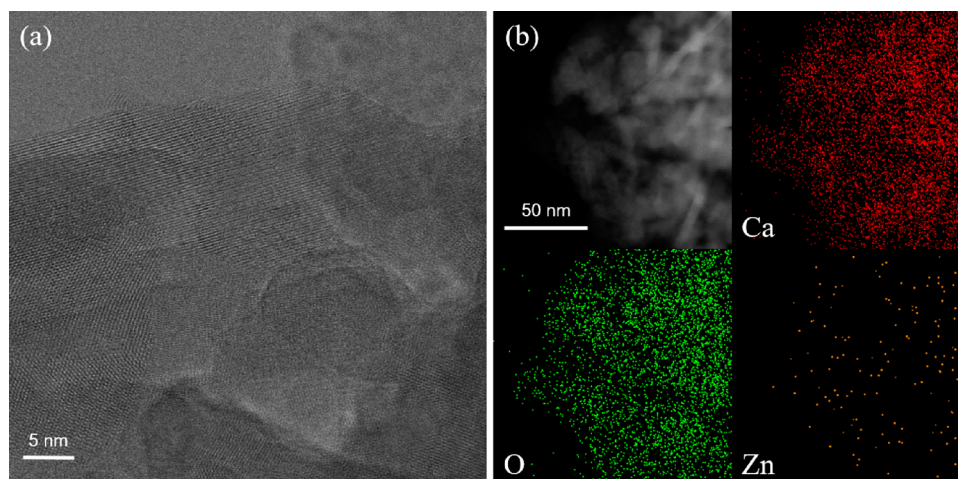


Figure 2. (a) STEM image and (b) EDS mapping of original, calcium, oxygen, and zinc of Zn-HAP (1.4 wt %).

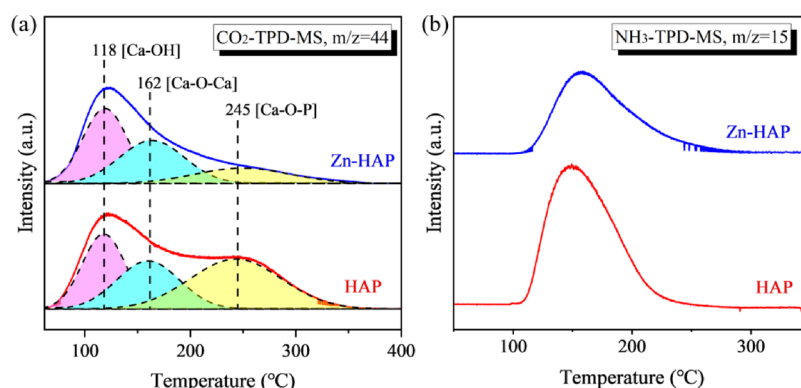


Figure 3. (a) CO₂-TPD-MS and (b) NH₃-TPD-MS profiles of HAP and Zn-HAP (1.4 wt %). Both samples were swept by a He flow for 1 h to remove weakly adsorbed gas before being heated.

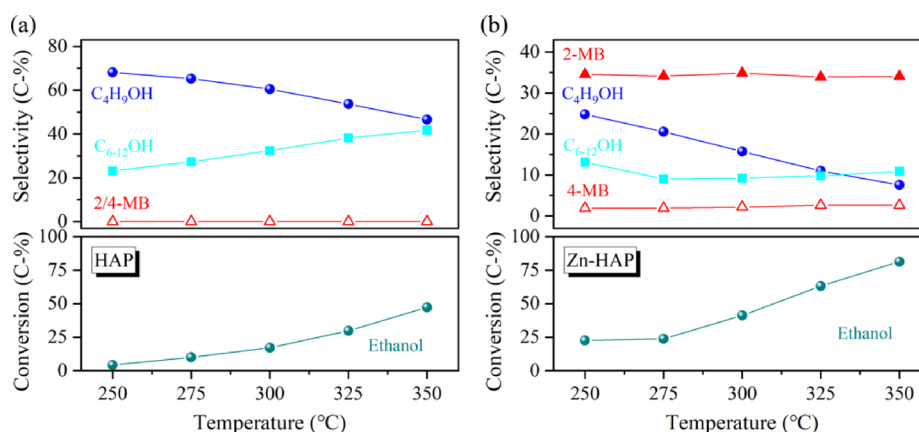


Figure 4. Temperature dependence of ethanol conversion and selectivity for major products over (a) HAP and (b) Zn-HAP (1.4 wt %). Reaction conditions: C₂H₅OH/N₂ = 6/94 and weight hourly space velocity (WHSV) = 1.1 gC₂H₅OH g_{cat}⁻¹ h⁻¹. In the diagram, 2-MB, 4-MB, C₄H₉OH, and C₆₋₁₂OH refer to ortho-aromatics, para-aromatics, *n*-butanol, and saturated aliphatic alcohols containing 6–12 carbon atoms, respectively.

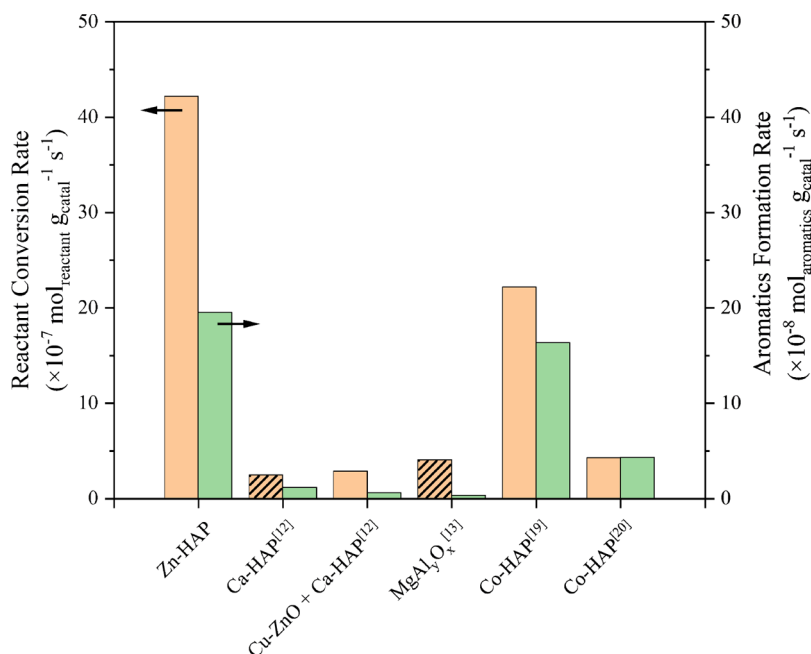


Figure 5. Comparison of the ethanol conversion rate (in orange) and aromatics formation rate (in green). The slash indicates that the reactant is acetaldehyde.

which implicates the facile formation of acetaldehyde that was faster than its consumption, i.e., the coupling of two

acetaldehyde molecules to produce 2-butenal can be considered as the reaction rate-determining step for upgrading

of ethanol to methylbenzyl alcohol. This is different from the previous work of ethanol conversion on Co-HAP, which considered the ethanol dehydrogenation as the rate-determining step given the low ethanol conversion (34.9%) and low presence of acetaldehyde (5.1% at 325 °C).¹⁹ The high excess presence of acetaldehyde (23.0%) and ethanol conversion (63.1% at 325 °C) on Zn-HAP suggest that the formation of acetaldehyde is no longer a limiting factor for the reaction. It is related to the dehydrogenation ability of the transition metals, indicating that the Zn site is a highly active site for ethanol dehydrogenation. The consumption of the nascent acetaldehyde, i.e., acetaldehyde coupling, occurs at basic sites on hydroxyapatite. We note that in a kinetic study of acetaldehyde conversion on HAP, the rate of aromatic oxide formation was shown to be directly proportional to the concentration of acetaldehyde,¹² implicating that acetaldehyde is a key intermediate to produce aromatics in the ethanol conversion catalyzed by HAP catalysts.

The [Ca–O–P] site has been proven to be responsible for the coupling reaction.³⁴ The substitution of the [Zn–O–P] sites for the [Ca–O–P] sites reduced the strong basic sites on the surface, resulting in the slowdown of the carbon chain growth. The consumption rate of ethanol is $4.22 \times 10^{-6} \text{ mol}_{\text{ethanol}} \text{ g}_{\text{catal}}^{-1} \text{ s}^{-1}$ and the formation rate of aromatic products is $1.95 \times 10^{-7} \text{ mol}_{\text{aromatics}} \text{ g}_{\text{catal}}^{-1} \text{ s}^{-1}$, both being much higher than previously reported (Figure 5).^{12,13,19,20} A scale-up synthesis experiment was performed with 10 g of Zn-HAP catalyst through the impregnation method, and similar catalytic activity and product distribution were obtained. This shows that the preparation of the catalyst does not require harsh conditions, showing that the catalyst has the prospect of large-scale applications.

3.2.2. Influence of Zn Loading. Table 1 shows the effect of Zn loading on the selectivity of the major oxygen-containing

Table 1. Loading Dependence of Ethanol Conversion and Selectivity of Major Products at 325 °C^a

loading (wt %)	Con. (%)	Sel. (%)				MB yield (%)
		CH ₃ CHO	C ₄ –OH	C _{6–12} –OH	MB	
HAP	29.8	1.6	53.6	38.2	0.4	0.1
0.7Zn-HAP	57.2	15.8	15.1	11.4	37.9	21.7
1.4Zn-HAP	63.1	23.0	11.0	9.8	36.5	23.0
2.7Zn-HAP	58.5	25.6	9.2	7.7	37.7	22.1
4.3Zn-HAP	54.5	22.5	12.0	7.9	37.9	20.6
5.9Zn-HAP	34.9	25.3	10.6	8.0	35.6	16.8

^aMB refers to the sum of ortho- and para-aromatics.

products. In general, the doping of Zn not only raised the conversion rate of ethanol but also significantly changed the product distribution. With the increase in Zn loading in the range of 0 to 1.4 wt %, the selectivity of aromatics increased sharply from 0.4 to 36.5%, and the selectivity of acetaldehyde gradually increased from 1.6 to 23.0%. Meanwhile, the selectivity of long-chain aliphatic alcohols, such as *n*-butanol, dramatically decreased from 53.6 to 11.0%. The results indicate that Zn dopants effectively promote ethanol dehydrogenation, thereby modulating the reaction path and the product distribution. The TOF value of Zn site could reach $2.51 \times 10^{-2} \text{ s}^{-1}$, which is much higher than that of the best previously reported transition metal-doped HAP catalyst (1.63

$\times 10^{-2} \text{ s}^{-1}$ on Co-HAP).¹⁹ When the Zn loading is above 1.4 wt %, the ethanol conversion decreases due to the growth of metal clusters (Figure S1) with poorer dehydrogenation activity (Figure S5).

3.2.3. Mechanism. The selectivity of aromatics on the Zn-HAP catalyst is ~ 100 times that on the HAP, implicating the different reaction pathways on the two catalysts. To understand the reasons for the significant difference between the two catalysts in catalyzing the ethanol conversion, experiments with various WHSV values were conducted to manipulate the contact time of ethanol on the Zn-HAP catalysts. Figure 6

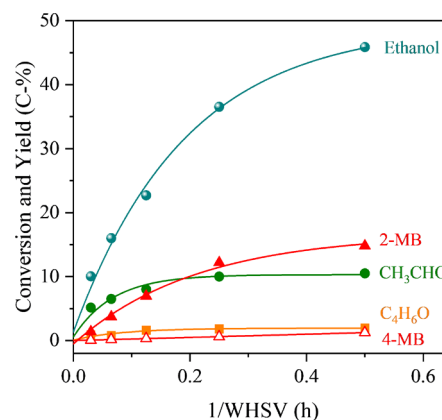


Figure 6. Dependence of the product distribution on the residence time (1/WHSV) over Zn-HAP (1.4 wt %) at 325 °C. Reaction conditions: C₂H₅OH/N₂ = 6/94, with the WHSV ranging from 2.0 to 33.4 gC₂H₅OH g_{cat}^{−1} h^{−1}.

shows the positive correlation between contact time, represented by 1/WHSV, and the yield of acetaldehyde on Zn-HAP, and the sharp increase at a short contact time suggests that acetaldehyde is the primary product formed by the ethanol dehydrogenation. As the contact time increases to above 0.2 h, the yield of acetaldehyde tends to be stable and the yield of 2-butenal remains low, indicating that 2-butenal can hardly accumulate under the experimental conditions and is rapidly converted to C₈ intermediates through a condensation or addition reaction, which is consistent with the increase in the rate of formation of 2-MB, and 2-MB becomes the main product. These results suggest that 2-butenal is an important intermediate, and its dimerization generates oxygen-containing aromatic compounds through C–C bond coupling and dehydrocyclization reactions.

To identify key C₈ intermediates controlling the selectivity of aromatics, 2-butenal was fed as a reactant. Figure 7 shows that both HAP and Zn-HAP catalyze the coupling of 2-butenal at 50 °C, with a 10% conversion on HAP and a 40% conversion on Zn-HAP, and 2-butenal is barely detected on both HAP and Zn-HAP, with its conversion marginally higher on the latter than on the former when the temperature is elevated to 300 °C. It implies that Zn sites can promote carbon chain growth. More than 95% of the C₈ enals are 2,4,6-octatrienal at 50 °C, which means that it is a precursor for 2-MB formation. More aromatics have been produced on Zn-HAP with a significantly larger quantity of methylbenzyl alcohols, indicating that the Zn site is also an active center for both cyclization and hydrogenation. At the same time, this confirms that dissociated hydrogen atoms from dehydrogenation can participate in the hydrogenation reaction. Either

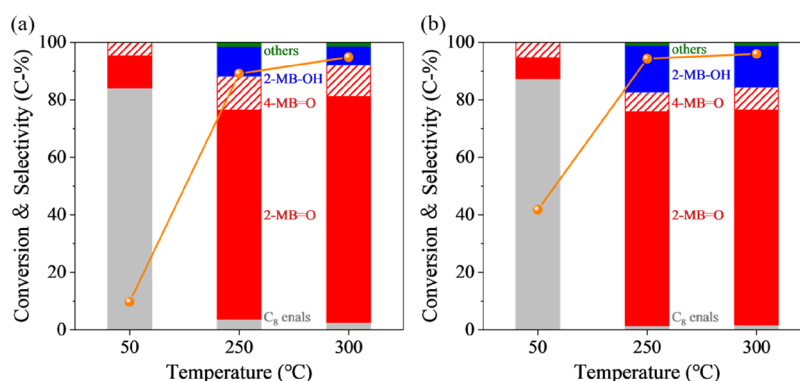
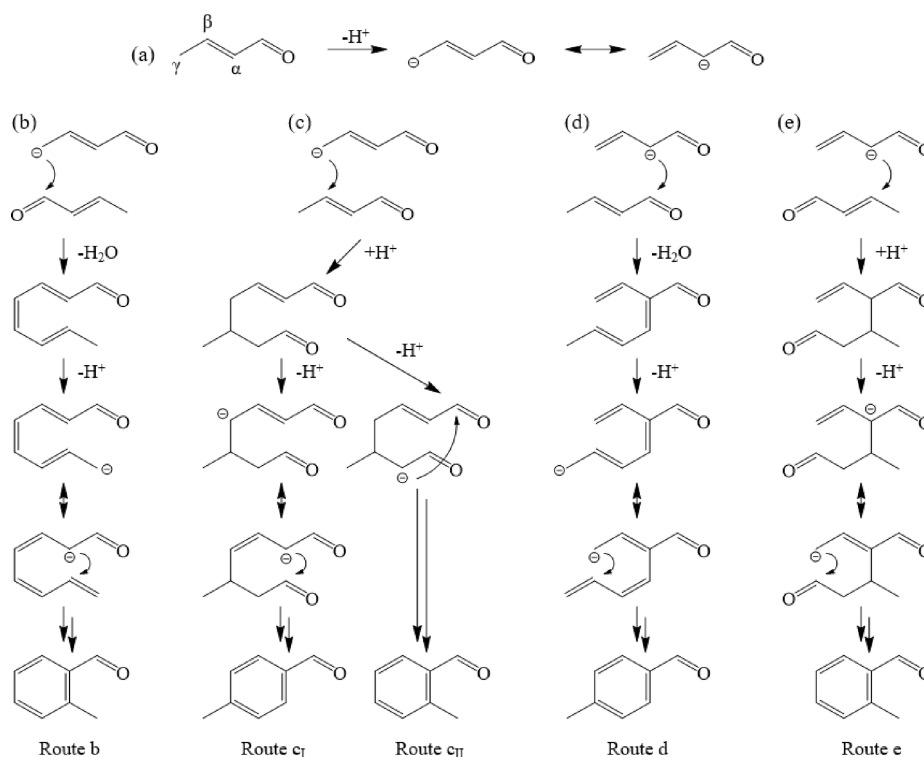


Figure 7. Temperature dependence of 2-butenal conversion and selectivity for major products over (a) HAP and (b) Zn-HAP (1.4 wt %). Reaction conditions: 2-butenal/N₂ = 0.5/99.5. In the diagram, C₈ enals, 2-MB=O, 4-MB=O, and 2-MB-OH refer to unsaturated aliphatic aldehydes containing eight carbon atoms, *ortho*-methylbenzaldehyde, *para*-methylbenzaldehyde, and *ortho*-methylbenzyl alcohol, respectively.

Scheme 1. (a) Deprotonation of 2-Butenal and Its Resonance Structure; (b–e) Four Routes of Aromatics Formation from 2-Butenal through Nucleophilic Attack



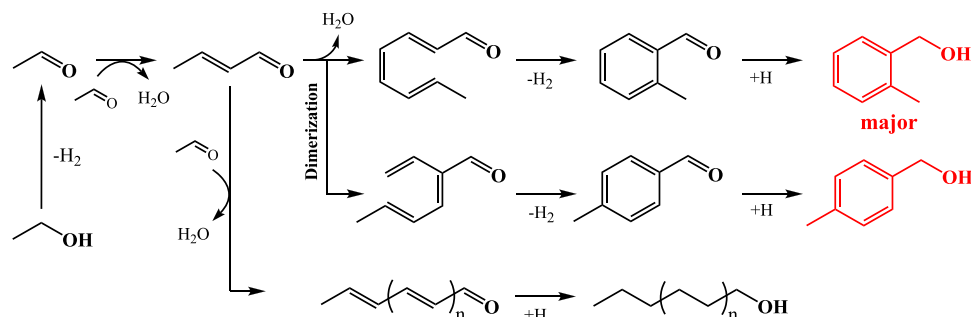
chain growth or cyclization involves the nucleophilic attack of a carbanion, which requires the cleavage of the C–H bond in prior. This suggests that the Zn sites can enhance the dehydrogenation.

Scheme 1 shows four routes of the nucleophilic reaction from 2-butenal to C₈ intermediates and finally to aromatics. The deprotonation of 2-butenal usually occurs at γ -C because γ -C carries a more negative charge than α -C and the negative charge of the sp² anion (α -C) cannot be stabilized, although this pathway is possible for the formation of 2-MB (Figure S6).^{13,35} The negative charge can be transferred to α -C via a conjugated π bond and form resonance structures (Scheme 1a). Similarly, the deprotonation of C₈ intermediates occurs at the sp³-C and transfers the charge to another carbon. As the carbonyl-C of 2-butenal is more electrophilic than the β -C,¹¹ the carbonyl-C is a preferential position to be attacked in the

condensation reaction to form a C=C bond, as demonstrated in Scheme 1 (routes b and d for the formation of *ortho*- and *para*-aromatics, respectively). Route b represents the main pathway, which is confirmed by the identification of the key intermediate 2,4,6-octatrienal (Figure 7). The deprotonation of 2,4,6-octatrienal at the terminal position is more difficult than that of 2-butenal, which determines that the cyclization will be slower than the coupling, making it difficult to cyclize enals at low temperatures (Figure 7).

Routes c and e are nucleophiles undergoing addition reactions at β -C to form a C–C bond. Although the rate of the addition reaction is not as high as that of the condensation reaction, the C₈ intermediates via the former contain two aldehyde groups, which are more likely to be deprotonated and cyclized. Route c_I is more likely than route c_{II} in view of the

Scheme 2. Proposed Pathways for Aromatic Products and Byproducts from Ethanol



relatively easier formation of carbanion in route c_I than in route c_{II} .¹¹

A similar pathway of cross addition reaction between 2-butenal and 2-butenol has been proposed.^{19,36} Butyraldehyde was detected in the 2-butenal feed test rather than 2-butenol, although the carbonyl group is more easily hydrogenated. This suggests the possibility that deprotonated 2-butenol can be consumed by a nucleophilic attack to 2-butenal. A Diels–Alder reaction between 2,4-hexadienal and ethylene to form para-aromatics is also proposed.³⁶ We note that in earlier study, no ethylene was detected in the ethanol conversion on HAP.⁶ In summary, the nucleophilic attack of the γ -C of protonated 2-butenal to the carbonyl-C of another 2-butenal and the subsequent cyclization constitute the main pathway of the ethanol upgrading on Zn-HAP, which is confirmed by the capture of the key intermediate 2,4,6-octatrienal.

The acetaldehyde maintains a high level presence and tops all the other nonaromatic products. In view of the yields of 2-butenal and acetaldehyde as a function of 1/WHSV (Figure 6), the consumption of acetaldehyde, i.e., its coupling to produce 2-butenal, is recognized as the rate-determining step of the reaction, and the step prior, i.e., the ethanol dehydrogenation, and the step posterior, i.e., the 2-butenal coupling, proceed relatively faster. The addition or condensation reactions between butenal and acetaldehyde or butenal form C_{6+} enals through nucleophilic attack. This allows the aldehyde group to be retained at the terminal position. The hydrogenation with dissociated hydrogen atoms as a H-donor or Meerwein–Ponndorf–Verley (MPV) reaction with ethanol can turn enal intermediates and methylbenzaldehyde into saturated alcohols.^{37–39} The mechanism of the ethanol upgrading to aromatics on Zn-HAP is summarized in Scheme 2.

3.2.4. Stability of the Catalyst. Stability is one of the key characteristic parameters in the evaluation of solid catalysts. Figure 8 shows the time evolution of the Zn-HAP catalyst performance in the conversion of ethanol to aromatic alcohol at 325 °C. During the initial stage of the reaction, the conversion rate declined first and then remained stable, and the 2-MB selectivity showed the same trend. The initial decrease in activity was attributed to the rapid polymerization of acetaldehyde into polycyclic substances covering part of the surface sites (Figure S7). This is confirmed by the recovery of the catalyst at 400 °C for 2 h in an air atmosphere.

The recovered catalyst was subjected to an ethanol conversion reaction under the same test conditions, and a similar catalytic performance to the previous catalytic test was obtained. This showed that the recovery of the catalyst can restore both its initial and operational activity as the fresh catalyst, and the conversion rate of ethanol and the selectivity

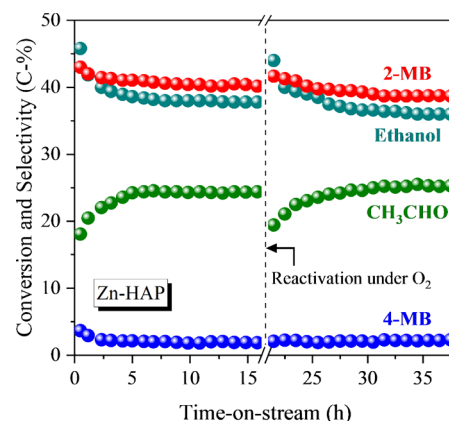


Figure 8. Stability test of Zn-HAP (1.4 wt %) in ethanol conversion. Reaction conditions: $C_2H_5OH/N_2 = 6/94$ and $WHSV = 2.2 \text{ g}_{C_2H_5OH} \text{ g}_{cat}^{-1} \text{ h}^{-1}$.

of aromatics were retained. These results show that the catalyst not only has good stability but also can be conveniently recovered with good recyclability. These properties enable it to be one of the potential industrial catalysts.

3.3. DFT Calculation. In the XRD characterization of Zn-HAP, the pattern of hexagonal hydroxyapatite was observed, which is also generally considered to be more active than monoclinic hydroxyapatite.⁴⁰ Thus, when the Zn-HAP models were prepared, the hexagonal conformation was adopted, as shown in Figure 9a. In the models, the calcium ions coordinated with nine oxygen atoms of phosphate groups are denoted as Ca_I sites, and those coordinated with six oxygen atoms of phosphate groups and one from the hydroxyl group are denoted as Ca_{II} sites.⁴⁰ Considering that no noticeable metal particles were observed at a low loading of Zn, when preparing the Zn-HAP model, only one Zn^{2+} cation was added to the HAP supercell to substitute for a Ca^{2+} ion at the Ca_I or Ca_{II} site, and the models are denoted as Zn_I -HAP (Figure 9b) and Zn_{II} -HAP (Figure 9c), respectively. The optimized lattice parameters of the three models are in good agreement with the previous calculated and experimental data (Table 2).

The structure of the catalyst determines its catalytic activity, and the catalyst undergoes a slight change in the lattice parameters while maintaining the crystallinity upon Zn doping at the Ca site on the HAP surface (Table 2). In the two doping types, the Zn_I ion maintains the balanced coordination configuration in the native HAP with all three Zn_I -O bond lengths around 1.93 Å, which is similar to Ca_I -O bond lengths around 2.39 Å in the native HAP, resulting in a slight change in lattice. The Zn_{II} atom could hardly maintain a coordination mode as in the native HAP before doping, and the bond

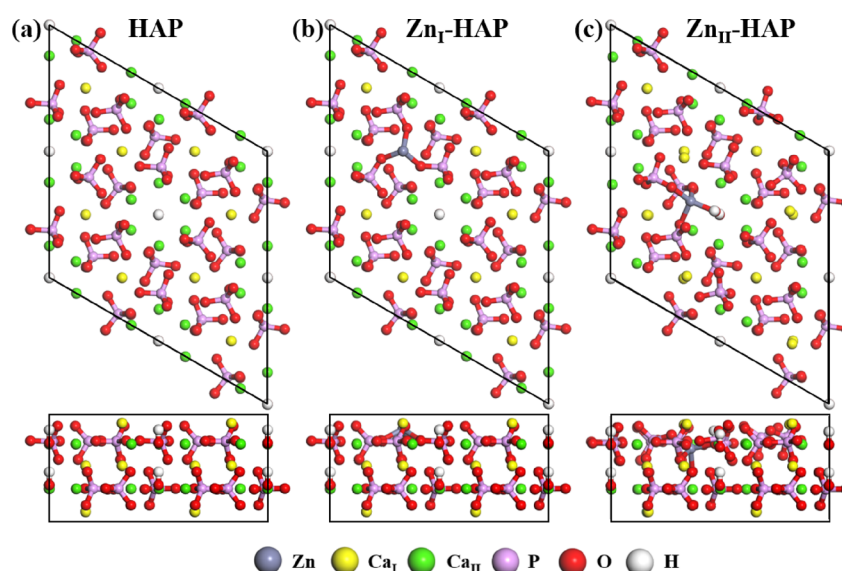


Figure 9. Top and side views of the optimized slab models: (a) HAP, (b) Zn_I-HAP, and (c) Zn_{II}-HAP. The 15 Å vacuum space is not shown in the side views for better visibility.

Table 2. Calculated and Experimentally Measured Lattice Parameters of HAP and Zn-HAP^a

system	<i>a</i> (Å)	<i>b</i> (Å)	<i>c</i> (Å)	ref.
HAP	9.45	9.45	6.88	this work
Zn _I -HAP	9.44	9.44	6.86	this work
Zn _{II} -HAP	9.42	9.45	6.87	this work
HAP ^{PBE}	9.55	9.55	6.94	41
HAP ^{LDA}	9.11	9.11	6.86	42
HAP ^{exptl}	9.43	9.43	6.89	43

^aThe calculated data of HAP and Zn-HAP in this work were converted to those of the unit cell for comparison with the experimental results.

lengths of the four Zn_{II}-O bonds are nearly equal (2.04, 2.02, and 2.08 Å) in contrast to the unequal values of Ca_{II}-O bond lengths before doping (2.27, 2.28, 2.33, and 2.43 Å). Besides, the hydroxyl group shifted toward the Zn_{II} atom and away from the center of the cell in Figure 9c, resulting in the lattice changes with an energy rise of 0.5 eV compared to the Zn_I-HAP model. It suggests that Zn_I is better suited to the coordination environment on the HAP surface.

On the other hand, the phosphate group covers part of the Ca_{II} site, which means steric effects for Zn_{II} doping. This is consistent with the TPD result, that is, the weak basic site [Ca-OH] remained essentially unchanged, while the population of [Ca-O-P] sites significantly reduced. The difference in coordination environment resulted in their distinct electronic structure, and the Zn_I carried less positive charge than the Zn_{II} (+1.24 lel vs +1.32 lel) according to the Bader charge analysis.

In the native HAP, the adsorption of ethanol on HAP relies on electrostatic interaction, e.g., the hydrogen bonding and the hole-lone pair (p- π) interaction. The doping of Zn changes the way of adsorption and activation of ethanol. The charge transfer (Figure 10) is responsible for the strong adsorption energy (Table S1) when ethanol adsorbed at the Zn_I site. The calculations also show that the Zn_I site plays a key role in the activation of the ethanol O-H bond in view of its elongated bond length from 0.97 to 1.05 Å upon adsorption. The energy

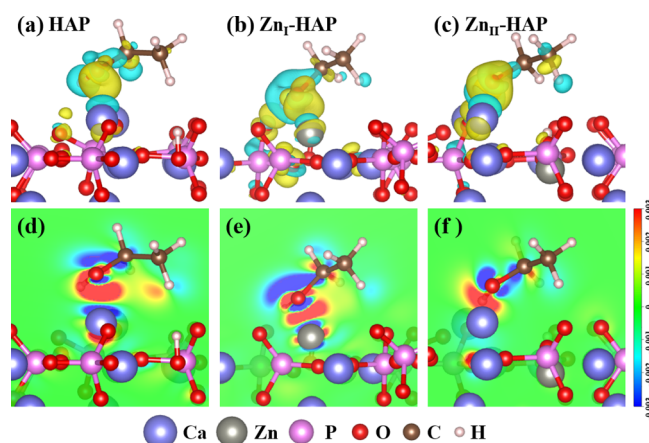


Figure 10. Charge transfer between the catalyst and the adsorbed ethanol: (a, d) HAP, (b, e) Zn_I-HAP, and (c, f) Zn_{II}-HAP. (a–c) Side views of the charge density difference upon the adsorption of ethanol (isovalue: 0.003 e/Bohr³, regions with electron accumulation in yellow and depletion in light blue) and (d–f) the corresponding plane traverse the M–O–H moiety (M = Ca for HAP and M = Zn for Zn-HAP).

barriers to the O–H bond cleavage of ethanol were calculated to be 0.66 and 0.14 eV at the Ca_I and Zn_I sites, respectively, indicating that the Zn_I site of Zn-HAP has significantly stronger ability in catalyzing the dehydrogenation of ethanol at its hydroxyl group than the Ca_I site in native HAP.

The dehydrogenation product acetaldehyde also showed a higher propensity to adsorb at the Zn_I site than on the other two sites. The strong adsorption at the Zn_I site can result in high coverage of acetaldehyde and promotes both its dehydrogenation reaction and the subsequent coupling reaction.

4. CONCLUSIONS

This work reported a bifunctional Zn-doped HAP catalyst that showed high activity to promote the direct conversion of ethanol to aromatic alcohols through dehydrogenation/aromatization. At 325 °C, the formation rate of aromatic

products is $1.95 \times 10^{-7} \text{ mol}_{\text{aromatics}} \text{ g}_{\text{catal}}^{-1} \text{ s}^{-1}$, which is the highest in the literature. The reaction pathway is green and sustainable as the feedstock ethanol is derived from renewable biomass resources, while the xylene route can be substituted to conserve petrochemical resources. The impregnation of Zn salts on HAP modified the amount of strong basic sites of the material and changed the activation mode of the ethanol feedstock, which first underwent a dehydrogenation reaction to obtain acetaldehyde, followed by a sequential condensation reaction to obtain the aromatics. During the reaction, the Zn sites displayed high dehydrogenation activity, and the accumulation of acetaldehyde, which was formed from the dehydrogenation of ethanol, suggested that its consumption via C–C coupling to produce 2-butanal was the rate-determining step of the reaction. This study provides insights into the development of novel strategies toward efficient direct synthesis of high value-added products from a sustainable resource.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.3c05085>.

XRD of Zn-HAP with different Zn loadings, in situ XRD and TEM of Zn-HAP, temperature dependence of ethanol conversion and selectivity with double ethanol feed, transition state of ethanol dehydrogenation, TG-MS of the spent catalyst, structural parameters and adsorption energy of ethanol and acetaldehyde, and reaction pathway of α -carbanion at the $\text{sp}^2\text{-C}$ (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Eagan, N. M.; Kumbhalkar, M. D.; Buchanan, J. S.; Dumesic, J. A.; Huber, G. W. Chemistries and Processes for the Conversion of Ethanol into Middle-Distillate Fuels. *Nat. Rev. Chem.* **2019**, *3*, 223–249.
- (2) Pérez Pimienta, J. A.; Papa, G.; Sun, J.; Stavila, V.; Sanchez, A.; Gladden, J. M.; Simmons, B. A. One-pot ethanol production under optimized pretreatment conditions using agave bagasse at high solids loading with low-cost biocompatible protic ionic liquid. *Green Chem.* **2022**, *24*, 207–217.
- (3) Pang, B.; Sun, Z. H.; Wang, L.; Chen, W. J.; Sun, Q.; Cao, X. F.; Shen, X. J.; Xiao, L.; Yan, J. L.; Deuss, P. J.; Yuan, T. Q.; Sun, R. C. Improved Value and Carbon Footprint by Complete Utilization of Corn cob Lignocellulose. *Chem. Eng. J.* **2021**, *419*, No. 129565.
- (4) Pang, J. F.; Yin, M.; Wu, P. F.; Li, X. Q.; Li, H. Y.; Zheng, M. Y.; Zhang, T. Advances in Catalytic Dehydrogenation of Ethanol to Acetaldehyde. *Green Chem.* **2021**, *23*, 7902–7916.
- (5) Wang, Q. N.; Shi, L.; Li, W.; Li, W. C.; Si, R.; Schuth, F.; Lu, A. H. Cu Supported on Thin Carbon Layer-Coated Porous SiO_2 for Efficient Ethanol Dehydrogenation. *Catal. Sci. Technol.* **2018**, *8*, 472–479.
- (6) Wang, Q. N.; Zhou, B. C.; Weng, X. F.; Lv, S. P.; Schuth, F.; Lu, A. H. Hydroxyapatite Nanowires Rich in [Ca–O–P] Sites for Ethanol Direct Coupling Showing High C_{6-12} Alcohol Yield. *Chem. Commun.* **2019**, *55*, 10420–10423.
- (7) Natte, K.; Dumrath, A.; Neumann, H.; Beller, M. Palladium-Catalyzed Carbonylations of Aryl Bromides using Paraformaldehyde: Synthesis of Aldehydes and Esters. *Angew. Chem., Int. Ed.* **2014**, *53*, 10090–10094.
- (8) Bohigues, B.; Rojas-Buzo, S.; Moliner, M.; Corma, A. Coordinatively Unsaturated Hf-MOF-808 Prepared via Hydrothermal Synthesis as a Bifunctional Catalyst for the Tandem N-Alkylation of Amines with Benzyl Alcohol. *ACS Sustainable Chem. Eng.* **2021**, *9*, 15793–15806.
- (9) Gaster, E.; Kozuch, S.; Pappo, D. Selective Aerobic Oxidation of Methylarenes to Benzaldehydes Catalyzed by N-Hydroxyphthalimide and Cobalt(II) Acetate in Hexafluoropropan-2-ol. *Angew. Chem., Int. Ed.* **2017**, *56*, 5912–5915.
- (10) Partenheimer, W. Methodology and Scope of Metal Bromide Autoxidation of Hydrocarbons. *Catal. Today* **1995**, *23*, 69–158.
- (11) Zhang, L.; Pham, T. N.; Faria, J.; Santhanaraj, D.; Sooknoi, T.; Tan, Q.; Zhao, Z.; Resasco, D. E. Synthesis of C_4 and C_8 Chemicals from Ethanol on MgO-Incorporated Faujasite Catalysts with Balanced Confinement Effects and Basicity. *ChemSusChem* **2016**, *9*, 736–748.
- (12) Moteki, T.; Rowley, A. T.; Flaherty, D. W. Self-Terminated Cascade Reactions That Produce Methylbenzaldehydes from Ethanol. *ACS Catal.* **2016**, *6*, 7278–7282.
- (13) Lusardi, M.; Struble, T.; Teixeira, A. R.; Jensen, K. F. Identifying the Roles of Acid-Base Sites in Formation Pathways of Tolualdehydes from Acetaldehyde over MgO-based Catalysts. *Catal. Sci. Technol.* **2020**, *10*, 536–548.

- (14) Hill, I. M.; Hanspal, S.; Young, Z. D.; Davis, R. J. DRIFTS of Probe Molecules Adsorbed on Magnesia, Zirconia, and Hydroxyapatite Catalysts. *J. Phys. Chem. C* **2015**, *119*, 9186–9197.
- (15) Diallo-Garcia, S.; Osman, M. B.; Krafft, J. M.; Casale, S.; Thomas, C.; Kubo, J.; Costentin, G. Identification of Surface Basic Sites and Acid-Base Pairs of Hydroxyapatite. *J. Phys. Chem. C* **2014**, *118*, 12744–12757.
- (16) Boukha, Z.; González-Prior, J.; Rivas, B. d.; González-Velasco, J. R.; López-Fonseca, R.; Gutiérrez-Ortiz, J. I. Synthesis, Characterisation and Behaviour of Co/Hydroxyapatite Catalysts in the Oxidation of 1,2-Dichloroethane. *Appl. Catal. B: Environ.* **2016**, *190*, 125–136.
- (17) Ho, C. R.; Defalque, V.; Zheng, S.; Bell, A. T. Propanol Amination over Supported Nickel Catalysts: Reaction Mechanism and Role of the Support. *ACS Catal.* **2019**, *9*, 2931–2939.
- (18) Low, H. R.; Avdeev, M.; Ramesh, K.; White, T. J. Zinc Hydroxyapatite Catalyst for Decomposition of 2-Propanol. *Adv. Mater.* **2012**, *24*, 4175–4179.
- (19) Wang, Q. N.; Weng, X. F.; Zhou, B. C.; Lv, S. P.; Miao, S.; Zhang, D.; Han, Y.; Scott, S. L.; Schüth, F.; Lu, A. H. Direct, Selective Production of Aromatic Alcohols from Ethanol Using a Tailored Bifunctional Cobalt-Hydroxyapatite Catalyst. *ACS Catal.* **2019**, *9*, 7204–7216.
- (20) Zhou, B.; Wang, Q.; Weng, X.; He, L.; Li, W.; Lu, A. Regulating Aromatic Alcohols Distributions by Cofeeding Methanol with Ethanol over Cobalt-Hydroxyapatite Catalyst. *ChemCatChem.* **2020**, *12*, 2341–2347.
- (21) Kresse, G.; Furthmüller, J. Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set. *Phys. Rev. B* **1996**, *54*, 11169–11186.
- (22) Kresse, G.; Furthmüller, J. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. *Comput. Mater. Sci.* **1996**, *6*, 15–50.
- (23) Blochl, P. E. Projector Augmented-Wave Method. *Phys. Rev. B* **1994**, *50*, 17953–17979.
- (24) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **1996**, *77*, 3865–3868.
- (25) Posner, A. S.; Perloff, A.; Diorio, A. F. Refinement of the Hydroxyapatite Structure. *Acta Crystallogr.* **1958**, *11*, 308–309.
- (26) Monkhorst, H. J.; Pack, J. D. Special Points for Brillouin-Zone Integrations. *Phys. Rev. B* **1976**, *13*, 5188–5192.
- (27) Grimme, S.; Ehrlich, S.; Goerigk, L. Effect of the Damping Function in Dispersion Corrected Density Functional Theory. *J. Comput. Chem.* **2011**, *32*, 1456–1465.
- (28) Henkelman, G.; Jonsson, H. A dimer method for finding saddle points on high dimensional potential surfaces using only first derivatives. *J. Chem. Phys.* **1999**, *111*, 7010–7022.
- (29) Bader, R. F. W. Atoms in Molecules. *Acc. Chem. Res.* **1985**, *18*, 9–15.
- (30) Henkelman, G.; Arnaldsson, A.; Jonsson, H. A fast and robust algorithm for Bader decomposition of charge density. *Comput. Mater. Sci.* **2006**, *36*, 354–360.
- (31) Jun, J. H.; Lee, T. J.; Lim, T. H.; Nam, S. W.; Hong, S. A.; Yoon, K. J. Nickel-calcium phosphate/hydroxyapatite catalysts for partial oxidation of methane to syngas: characterization and activation. *J. Catal.* **2004**, *221*, 178–190.
- (32) Sugiyama, S.; Shono, T.; Makino, D.; Moriga, T.; Hayashi, H. Enhancement of the catalytic activities in propane oxidation and H-D exchangeability of hydroxyl groups by the incorporation with cobalt into strontium hydroxyapatite. *J. Catal.* **2003**, *214*, 8–14.
- (33) TSUCHIDA, T.; KUBO, J.; YOSHIOKA, T.; SAKUMA, S.; TAKEGUCHI, T.; UEDA, W. Reaction of ethanol over hydroxyapatite affected by Ca/P ratio of catalyst. *J. Catal.* **2008**, *259*, 183–189.
- (34) Ho, C. R.; Shylesh, S.; Bell, A. T. Mechanism and Kinetics of Ethanol Coupling to Butanol over Hydroxyapatite. *ACS Catal.* **2016**, *6*, 939–948.
- (35) Iriarte, I.; Olaizola, O.; Vera, S.; Gamboa, I.; Oiarbide, M.; Palomo, C. Controlling the alpha/gamma-Reactivity of Vinylous Ketone Enolates in Organocatalytic Enantioselective Michael Reactions. *Angew. Chem., Int. Ed.* **2017**, *56*, 8860–8864.
- (36) Moteki, T.; Rowley, A. T.; Bregante, D. T.; Flaherty, D. W. Formation Pathways toward 2- and 4-Methylbenzaldehyde via Sequential Reactions from Acetaldehyde over Hydroxyapatite Catalyst. *ChemCatChem.* **2017**, *9*, 1921–1929.
- (37) Young, Z. D.; Davis, R. J. Hydrogen transfer reactions relevant to Guerbet coupling of alcohols over hydroxyapatite and magnesium oxide catalysts. *Catal. Sci. Technol.* **2018**, *8*, 1722–1729.
- (38) Moteki, T.; Flaherty, D. W. Mechanistic Insight to C-C Bond Formation and Predictive Models for Cascade Reactions among Alcohols on Ca- and Sr-Hydroxyapatites. *ACS Catal.* **2016**, *6*, 4170–4183.
- (39) Ogo, S.; Onda, A.; Yanagisawa, K. Selective synthesis of 1-butanol from ethanol over strontium phosphate hydroxyapatite catalysts. *Appl. Catal. A-Gen* **2011**, *402*, 188–195.
- (40) Oh, S. C.; Xu, J.; Tran, D. T.; Liu, B.; Liu, D. Effects of Controlled Crystalline Surface of Hydroxyapatite on Methane Oxidation Reactions. *ACS Catal.* **2018**, *8*, 4493–4507.
- (41) Zilm, M. E.; Chen, L.; Sharma, V.; McDannald, A.; Jain, M.; Ramprasad, R.; Wei, M. Hydroxyapatite substituted by transition metals: experiment and theory. *Phys. Chem. Chem. Phys.* **2016**, *18*, 16457–16465.
- (42) Calderin, L.; Stott, M. J.; Rubio, A. Electronic and crystallographic structure of apatites. *Phys. Rev. B* **2003**, *67*, No. 134106.
- (43) Kim, J. Y.; Fenton, R. R.; Hunter, B. A.; Kennedy, B. J. Powder diffraction studies of synthetic calcium and lead apatites. *Aust. J. Chem.* **2000**, *53*, 679–686.